Politecnico di Milano

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M.Sc. in Aeronautical Engineering

ASSIGNMENT 1

Structural Dynamics and Aeroelasticity

Aeroelastic Analysis and Optimization of a Braced Wing Configuration

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Abstract

This report details the aeroelastic characterization of a cantilever wing reinforced by a structural strut. Utilizing a Ritz-Galerkin numerical approach, the study evaluates the divergence dynamic pressure, control effectiveness, and reversal speed. Furthermore, a structural optimization is performed to expand the aeroelastic envelope while minimizing weight penalties, demonstrating the critical role of the strut's geometric configuration.

1 Introduction and Theoretical Framework

This research investigates the static aeroelastic response of a cantilevered wing model, structurally enhanced by an elastic strut integrated at mid-span. The analysis aims to quantify the impact of this braced configuration on the system's static stability and control authority.

1.1 Problem Modeling

The wing is idealized as a beam undergoing coupled bending and torsion. The kinematic fields, defined by the vertical displacement $w(y)$ and the twist angle $\theta(y)$, are approximated via the Ritz-Galerkin method:

$$w(y) \approx \sum_{i=1}^{N_w} q_{w,i}\phi_i(y), \quad \theta(y) \approx \sum_{j=1}^{N_\theta} q_{\theta,j}\psi_j(y) \tag{1}$$

Adopting the cantilever boundary conditions at the root:

$$\begin{cases} w(0) = 0 \\ w'(0) = 0 \\ \theta(0) = 0 \end{cases} \tag{2}$$

the trial functions are selected as $\phi(y) = (y/b)^2$ and $\psi(y) = (y/b)$. The corresponding curvature and twist rate are expressed as: $w'' = \frac{2}{b^2}q_w$ and $\theta' = \frac{1}{b}q_\theta$.

1.2 Principle of Virtual Work (PVW)

The governing equilibrium equations are derived from the Principle of Virtual Work ($\delta W_{int} = \delta W_{ext}$). The internal virtual work accounts for the wing's strain energy and the supplemental elastic contribution of the strut:

$$\delta W_i = \int_0^b \delta w''^T EJ w'' \, dy + \int_0^b \delta\theta'^T GJ\theta' \, dy + \delta w_B^T k_{strut} w_B \tag{3}$$

where $w_B = w(b/2) - x_B\theta(b/2)$ represents the vertical displacement at the strut attachment point B. The negative sign associated with the chordwise offset x_B arises from the standard aeronautical convention, where a nose-up pitching moment is defined as positive; consequently, points located aft of the elastic axis exhibit a negative displacement for a positive rotation θ. The equivalent axial stiffness of the strut is defined as $k_{strut} = \frac{EA}{L_{strut}}\sin^2(\gamma)$.

The external virtual work comprises the aerodynamic loads generated by both the wing profile and the aileron deflection:

$$\delta W_e = \int_0^b \delta w_{AC}^T qcC_{L\alpha}\theta(y) \, dy + \int_{b-b_a}^b \delta w_{AC}^T qcC_{L\beta}\beta \, dy + \int_{b-b_a}^b \delta\theta^T qc^2 C_{MAC\beta}\beta \, dy \tag{4}$$

where $w_{AC} = w(y) + e\theta(y)$ denotes the displacement at the aerodynamic center.

1.3 System Governing Equations

Upon discretization, the static equilibrium of the aeroelastic system is governed by the following matrix equation:

$$\begin{Bmatrix} \delta q_w \\ \delta q_\theta \end{Bmatrix}^T (\mathbf{K}_1 + \mathbf{K}_2) \begin{Bmatrix} q_w \\ q_\theta \end{Bmatrix} = \begin{Bmatrix} \delta q_w \\ \delta q_\theta \end{Bmatrix}^T \left(q\mathbf{K}_A \begin{Bmatrix} q_w \\ q_\theta \end{Bmatrix} + \mathbf{F}_\beta \right) \tag{5}$$

2 Divergence Dynamic Pressure Analysis

The aeroelastic stability of the system is governed by the spectral properties of the total stiffness operator. To determine the divergence dynamic pressure, a static equilibrium problem is formulated in matrix form, integrating the intrinsic structural stiffness of the wing, the stiffening effect of the strut, and the induced aerodynamic loads.

The stiffness matrices for the 2-D.o.F. model, accounting for bending and torsional degrees of freedom, are defined as follows:

- **Wing Structural Stiffness Matrix ($\mathbf{K}_1$):** Represents the inherent elastic restoration of the cantilevered wing structure.
- **Strut Stiffness Matrix ($\mathbf{K}_2$):** Models the supplemental stiffness introduced by the axial constraint as a function of its chordwise attachment point x_B.
- **Aerodynamic Influence Matrix ($\mathbf{K}_A$):** Describes the aerodynamic lift and moment resulting from the elastic deformation of the lifting surface.

The constituent matrices take the following form:

$$\mathbf{K}_1 = \begin{bmatrix} \frac{4EJ}{b^3} & 0 \\ 0 & \frac{GJ}{b} \end{bmatrix}, \quad \mathbf{K}_2 = \begin{bmatrix} \frac{k}{16} & -\frac{kx_B}{8} \\ -\frac{kx_B}{8} & \frac{kx_B^2}{4} \end{bmatrix}, \quad \mathbf{K}_A = \begin{bmatrix} 0 & cC_{L\alpha}\frac{b}{4} \\ 0 & ceC_{L\alpha}\frac{b}{3} \end{bmatrix} \tag{6}$$

The divergence dynamic pressure q_D is identified by the criticality condition where the determinant of the total system stiffness matrix vanishes, indicating the onset of static aeroelastic instability:

$$\det(\mathbf{K}_1 + \mathbf{K}_2 - q\mathbf{K}_A) = 0 \tag{7}$$

By solving the characteristic equation, the analytical expression for the divergence dynamic pressure is derived as a function of the chordwise position of point B:

$$q_D(x_B) = \frac{6\left(16(4\,EJ\,GJ) + 16\,b(EJ\,k\,x_B^2) + k\,GJ\,b^3\right)}{b\,(32(4\,EJ\,c\,C_{L\alpha}\,e\,b) + 2b^3(kcC_{L\alpha}eb) + 3b^3(c\,C_{L\alpha}\,b\,k\,x_B))} \tag{8}$$

3 Control Effectiveness and Strut Sensitivity Analysis

The control effectiveness, E_c, is a fundamental parameter in aeroelasticity used to quantify the influence of structural flexibility on the aerodynamic authority of control surfaces. It is analytically defined as the ratio between the total lift produced by the flexible wing and the lift generated by a theoretically rigid structure:

$$E_c = \frac{L_{total}}{L_R} = 1 + \frac{L_\theta}{L_R} \tag{9}$$

3.1 Aerodynamic Loading and Equilibrium

The rigid lift contribution, L_R, is a function of the dynamic pressure q, the chord c, the aileron lift coefficient $C_{L\beta}$, and the deflection angle β_0 over the aileron span b_a:

$$L_R = qcC_{L\beta}b_a\beta_0 \tag{10}$$

The elastic lift increment, L_θ, arises from the spanwise torsional deformation $\theta(y)$. By integrating the local lift contribution induced by this twist along the span b, we obtain:

$$L_\theta = \int_0^b qcC_{L\alpha}\theta(y)dy = qcC_{L\alpha}q_\theta\frac{b}{2} \tag{11}$$

To determine the generalized coordinates for bending (q_w) and torsion (q_θ), the coupled aeroelastic equilibrium system is solved using the Ritz-Galerkin approach:

$$\begin{Bmatrix} q_w \\ q_\theta \end{Bmatrix} = (\mathbf{K}_1 + \mathbf{K}_2 - q\mathbf{K}_A)^{-1}\mathbf{F}_\beta \tag{12}$$

Where $\mathbf{K}_1$ represents the wing stiffness, $\mathbf{K}_2$ the strut contribution, and $\mathbf{K}_A$ the aerodynamic influence matrix. The aerodynamic force vector $\mathbf{F}_\beta$ accounts for both the lift and the pitching moment generated by the aileron deflection:

$$\mathbf{F}_\beta = q\beta \begin{Bmatrix} cC_{L\beta}\frac{b^3-(b-b_a)^3}{3b^2} \\ c^2C_{mAC\beta}\frac{b^2-(b-b_a)^2}{2b} + ceC_{L\beta}\frac{b^2-(b-b_a)^2}{2b} \end{Bmatrix} \tag{13}$$

3.2 Parametric Sensitivity to Strut Geometry

To evaluate how the structural configuration influences the aeroelastic response, a sensitivity analysis is performed by varying the strut angle γ within the range $[10°, 20°]$. The ground attachment point A is assumed fixed, implying a constant vertical offset z_A:

$$z_A = \frac{b}{2}\tan(\gamma_{baseline}) \tag{14}$$

For any variation in γ, the geometric parameters of the strutnamely the length L_{strut}, the spanwise attachment coordinate y_B, and the effective vertical stiffness k_{new}must be updated:

$$L_{strut} = \frac{z_A}{\sin(\gamma_{new})}, \quad y_B = \frac{z_A}{\tan(\gamma_{new})}, \quad k_{new} = \frac{EA}{L_{strut}}\sin^2(\gamma_{new}) \tag{15}$$

The contribution of the strut to the global stiffness matrix, $\mathbf{K}_2$, is consequently modified to reflect the new attachment location y_B:

$$\mathbf{K}_2 = k_{new}\begin{bmatrix} (y_B/b)^4 & -x_B(y_B/b)^3 \\ -x_B(y_B/b)^3 & x_B^2(y_B/b)^2 \end{bmatrix} \tag{16}$$

3.3 Computational Considerations

By solving the system across a range of dynamic pressures q, the evolution of E_c is mapped for different strut angles γ, maintaining the chordwise position fixed at $x_B = -0.1$. It is important to observe that since both the elastic and rigid lift components are linearly dependent on the aileron deflection, the term β_0 cancels out in the ratio E_c. Consequently, the control effectiveness remains independent of the specific magnitude of the control input, representing an intrinsic property of the aeroelastic system.

4 Deformed Shape Comparison

This section presents the comparative analysis between the baseline cantilever wing and the braced configuration to evaluate the structural impact of the redundant constraint. The spanwise displacement and twist distributions are reconstructed at a reference dynamic pressure $q = 0.4q_{D,clean}$.

4.1 Reference Divergence Condition

The reference pressure $q_{D,clean}$ corresponds to the divergence dynamic pressure of the unbraced (clean) wing. In this configuration, the strut contribution vanishes ($\mathbf{K}_2 = 0$). The divergence condition is identified by the singularity of the aeroelastic stiffness matrix:

$$\det(\mathbf{K}_1 - q\mathbf{K}_A) = 0 \tag{17}$$

Assuming a single-degree-of-freedom Ritz expansion for torsion, the divergence dynamic pressure is defined as:

$$q_{D,clean} = \frac{3\,GJ}{c\,e\,b^2\,C_{L\alpha}} \tag{18}$$

4.2 Aeroelastic Equilibrium

The quasi-static aeroelastic response is determined by solving the system of equations for the generalized coordinates. The analysis considers a fixed aileron deflection $\beta_0 = 10°$. The generalized coordinates for the **braced configuration** ($x_B = -0.1$, $y_B = b/2$) are retrieved by inverting the augmented stiffness matrix:

$$\begin{Bmatrix} q_w \\ q_\theta \end{Bmatrix}_{braced} = (\mathbf{K}_1 + \mathbf{K}_2 - q\mathbf{K}_A)^{-1}\mathbf{F}_\beta \tag{19}$$

where $\mathbf{K}_2$ represents the stiffness contribution of the strut. For the **unbraced configuration**, the equilibrium simplifies to:

$$\begin{Bmatrix} q_w \\ q_\theta \end{Bmatrix}_{clean} = (\mathbf{K}_1 - q\mathbf{K}_A)^{-1}\mathbf{F}_\beta \tag{20}$$

4.3 Displacement Field Reconstruction

The physical deformation fields, representing the vertical translation $w(y)$ and the torsional rotation $\theta(y)$, are reconstructed by superimposing the chosen Ritz shape functions over the wingspan:

$$w(y) = \left(\frac{y}{b}\right)^2 q_w, \quad \theta(y) = \frac{y}{b} q_\theta \tag{21}$$

This formulation allows for a direct comparison of the spanwise deformation profiles between the two structural architectures, highlighting the effectiveness of the strut in mitigating aeroelastic deformations at sub-critical dynamic pressures.

5 Control Reversal Analysis

Control reversal occurs when the aerodynamic lift increment produced by an aileron deflection β is exactly offset by the lift reduction caused by the resulting aeroelastic twist of the wing. This condition is defined by the vanishing of the net lift increment ($\Delta L = 0$):

$$L = L_\beta + L_\theta = qcC_{L\beta}b_a\beta + qcC_{L\alpha}\int_0^b \theta(y)dy = qcC_{L\beta}b_a\beta + qcC_{L\alpha}q_\theta\frac{b}{2} = 0 \tag{22}$$

From this equation we can obtain the relationship between the control input β and the generalized torsional coordinate q_θ is derived:

$$\beta = -\frac{C_{L\alpha}\, b}{2\, C_{L\beta} b_a} q_\theta = -5.6\, q_\theta \tag{23}$$

The generalized aerodynamic forcing vector, calculated numerically for the first modes, is given by:

$$\underline{F}_\beta = q \begin{Bmatrix} 6.535 \\ 1.5605 \end{Bmatrix} \beta \tag{24}$$

To determine the reversal dynamic pressure q_R, the condition $\beta(q_\theta)$ is substituted back into the governing aeroelastic equations. This transforms the non-homogeneous forcing term into a state-dependent term, effectively modifying the aerodynamic influence matrix $\underline{\underline{K_A}}$ into a "reversal" matrix $\underline{\underline{K_A}}_R$:

$$q\underline{\underline{K_A}}_R = q \begin{bmatrix} 0 & \left(cC_{L\alpha}\frac{b}{4} - 5.6 \cdot 6.535\right) \\ 0 & \left(cC_{L\alpha}e\frac{b}{3} - 5.6 \cdot 1.5605\right) \end{bmatrix} \tag{25}$$

Unbraced Configuration In the cantilever (clean) case, the structural coupling matrix $\underline{\underline{K}}_2$ is null. The equations decouple, allowing for a direct analytical solution of the second kinematic equation:

$$GJ\, q_\theta = q \left(cC_{L\alpha} e \frac{b}{3} - 5.6 \cdot 1.5605 \right) q_\theta \tag{26}$$

Solving for q yields the baseline reversal pressure:

$$q_{R,clean} = \frac{GJ/b}{cC_{L\alpha}e\frac{b}{3} - 5.6 \cdot 1.5605} \tag{27}$$

Braced Configuration For the braced wing, the bending and torsional degrees of freedom are structurally coupled by the strut stiffness matrix $\underline{\underline{K}}_2(x_B)$. The reversal dynamic pressure q_R is therefore identified by solving the generalized eigenvalue problem:

$$\det\left[\left(\underline{\underline{K}}_1 + \underline{\underline{K}}_2(x_B)\right) - q\underline{\underline{K_A}}_R\right] = 0 \tag{28}$$

This analysis is repeated for varying chordwise positions x_B.

6 Numerical Convergence and Ritz-Galerkin Formulation

The reliability of the numerical results is strictly dependent on the convergence of the Ritz-Galerkin discretization. The continuous displacement fields for bending (w) and torsion (θ) are approximated by a finite expansion of N shape functions. The selected polynomial basis satisfies the essential boundary conditions at the wing root ($y = 0$):

$$\begin{cases} \phi_i(y) = \left(\frac{y}{b}\right)^{i+1} & \implies \phi_i(0) = 0, \phi_i'(0) = 0 \\ \psi_i(y) = \left(\frac{y}{b}\right)^{i} & \implies \psi_i(0) = 0 \end{cases} \tag{29}$$

$$\begin{cases} \phi_i''(y) = \dfrac{(i+1)i}{b^2}\left(\dfrac{y}{b}\right)^{i-1} \\ \psi_i'(y) = \dfrac{i}{b}\left(\dfrac{y}{b}\right)^{i-1} \end{cases} \tag{30}$$

The internal virtual work δW_i and the external virtual work δW_e are formulated to derive the generalized stiffness and aerodynamic matrices. The internal virtual work accounts for both

the wing's elastic strain energy and the discrete contribution of the strut at the attachment point y_B:

$$\delta W_i = \begin{Bmatrix} \delta \underline{q}_w \\ \delta \underline{q}_\theta \end{Bmatrix}^T \left(\underbrace{\begin{bmatrix} EJ \int_0^b \underline{\phi}''^T \underline{\phi}'' dy & \underline{\underline{0}} \\ \underline{\underline{0}} & GJ \int_0^b \underline{\psi}'^T \underline{\psi}' dy \end{bmatrix}}_{\underline{\underline{K}}_1} + \underbrace{\begin{bmatrix} k_s \underline{\phi}^T(y_B)\underline{\phi}(y_B) & -x_B k_s \underline{\phi}^T(y_B)\underline{\psi}(y_B) \\ -x_B k_s \underline{\psi}^T(y_B)\underline{\phi}(y_B) & x_B^2 k_s \underline{\psi}^T(y_B)\underline{\psi}(y_B) \end{bmatrix}}_{\underline{\underline{K}}_2} \right) \begin{Bmatrix} \underline{q}_w \\ \underline{q}_\theta \end{Bmatrix} \tag{31}$$

The external virtual work δW_e incorporates the non-conservative aerodynamic loads and the aileron forcing terms:

$$\delta W_e = \begin{Bmatrix} \delta \underline{q}_w \\ \delta \underline{q}_\theta \end{Bmatrix}^T \left(q \underbrace{\begin{bmatrix} \underline{\underline{0}} & cC_{L/\alpha} \int_0^b \underline{\phi}^T \underline{\psi} dy \\ \underline{\underline{0}} & ceC_{L/\alpha} \int_0^b \underline{\psi}^T \underline{\psi} dy \end{bmatrix}}_{\underline{\underline{K_A}}} \begin{Bmatrix} \underline{q}_w \\ \underline{q}_\theta \end{Bmatrix} + q\beta \underbrace{\begin{Bmatrix} cC_{L/\beta} \int_{b-b_a}^b \underline{\phi}^T dy \\ c^2 C_{mAC/\beta} \int_{b-b_a}^b \underline{\psi}^T dy + ceC_{L/\beta} \int_{b-b_a}^b \underline{\psi}^T dy \end{Bmatrix}}_{\underline{Q}_\beta} \right) \tag{32}$$

By fixing the strut attachment at $y_B = b/2$ and $x_B = -0.1$ m, the divergence dynamic pressure q_D is computed by solving the generalized eigenvalue problem:

$$\det \left(\underline{\underline{K}}_{total} - q \underline{\underline{K_A}} \right) = 0 \tag{33}$$

The convergence is assessed by iteratively increasing the number of shape functions N.

7 Design Optimization and Mass Minimization

The final objective of this study is to satisfy the design constraint $q_D^{new} \geq 1.4 q_D^{baseline}$ while minimizing the structural mass penalty. To achieve this, the optimization strategy focuses on the geometric reconfiguration of the strut rather than modifying the material properties or the cross-sectional area of the components, which would lead to a direct and significant mass increase.

The design variable selected for the optimization is the strut inclination angle γ. To maintain a consistent comparison, the vertical distance of the strut root attachment from the wing plane, z_A, is kept constant and defined by the baseline geometry as $z_A = (b/2)\tan(\gamma_{old})$. Consequently, a variation in γ dictates both the new spanwise attachment point y_B and the updated effective vertical stiffness k_{new}, according to the following analytical relations:

$$k_{new} = \frac{EA}{L_{new}} \sin^2(\gamma_{new}), \quad y_B = \frac{z_A}{\tan(\gamma_{new})}, \quad L_{new} = \frac{z_A}{\sin(\gamma_{new})} \tag{34}$$

By performing a parametric sweep of γ, the stability of the system was evaluated through the analysis of the minimum eigenvalue of the aeroelastic matrix. This approach allowed for the identification of an optimal γ that ensures the required 40% increase in divergence pressure. This geometric optimization leverages the coupling between the bracing stiffness and the spanwise load distribution, achieving the stability target through enhanced structural efficiency rather than a brute-force increase in material volume.